MAT 1272

Measures of Position and Box-and-Whisker Plots

Today's Goals

By the end of this lesson, students should be able to:

1. Understand measures of position.
2. Find quartiles Q_1, Q_2, and Q_3.
3. Compute the interquartile range.
4. Understand percentiles and percentile rank.
5. Construct a box-and-whisker plot.
6. Identify possible outliers using inner fences.
7. Describe the shape of a distribution using a boxplot.

Definition

A **measure of position** describes where a value is located relative to the other values in a data set.

Examples of measures of position:

- Quartiles
- Percentiles
- Percentile ranks

Class Discussion

If two students both score 80 on an exam, did they necessarily perform equally well compared with their classmates?

Why Position Matters

Suppose two students both score 80.

Student	Score	Class Information
A	80	Better than 40% of the class
B	80	Better than 90% of the class

Key Idea

The same data value can have a very different meaning depending on its position in the data set.

Quartiles

Definition

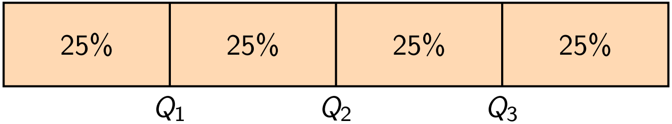
Quartiles divide a ranked data set into four equal parts.

The three quartiles are:

$$Q_1, \qquad Q_2, \qquad Q_3$$

- Q_1 is the first quartile.
- Q_2 is the second quartile, also called the median.
- Q_3 is the third quartile.

Visual Meaning of Quartiles



Interpretation

- About 25% of the data are below Q_1.
- About 50% of the data are below Q_2.
- About 75% of the data are below Q_3.

Example: Commuting Times

A sample of 12 students reported their one-way commuting times in minutes.

$$29,\ 14,\ 39,\ 17,\ 7,\ 47,\ 63,\ 37,\ 42,\ 18,\ 24,\ 55$$

Question

What should we always do first before finding quartiles?

Answer

Rank the data from smallest to largest.

Step 1: Rank the Data

Original data:

$$29,\ 14,\ 39,\ 17,\ 7,\ 47,\ 63,\ 37,\ 42,\ 18,\ 24,\ 55$$

Ranked data:

$$7,\ 14,\ 17,\ 18,\ 24,\ 29,\ 37,\ 39,\ 42,\ 47,\ 55,\ 63$$

Important

Quartiles are found using the **ranked** data set.

Step 2: Find the Median Q_2

Ranked data:

$$7,\ 14,\ 17,\ 18,\ 24,\ 29,\ 37,\ 39,\ 42,\ 47,\ 55,\ 63$$

There are 12 values, so the median is the average of the 6th and 7th values.

$$Q_2 = \frac{29 + 37}{2}$$

$$Q_2 = \frac{66}{2} = 33$$

Conclusion

$$Q_2 = 33$$

Step 3: Find Q_1

The values smaller than the median are:

$$7,\ 14,\ 17,\ 18,\ 24,\ 29$$

The first quartile is the median of this lower half.

$$Q_1 = \frac{17+18}{2}$$

$$Q_1 = 17.5$$

Conclusion

$$Q_1 = 17.5$$

Step 4: Find Q_3

The values larger than the median are:

$$37, \ 39, \ 42, \ 47, \ 55, \ 63$$

The third quartile is the median of this upper half.

$$Q_3 = \frac{42 + 47}{2}$$

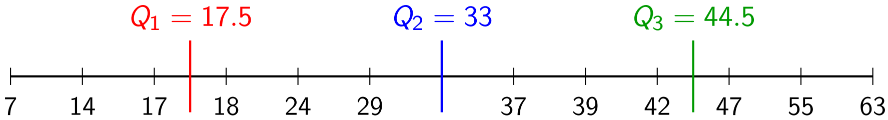
$$Q_3 = 44.5$$

Conclusion

$$Q_3 = 44.5$$

Quartiles Summary

7, 14, 17, 18, 24, 29, 37, 39, 42, 47, 55, 63



Interquartile Range

Definition

The **interquartile range**, or IQR, measures the spread of the middle 50% of the data.

Formula:

$$IQR = Q_3 - Q_1$$

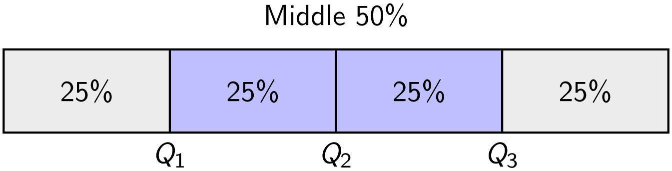
For the commuting-time example:

$$IQR = 44.5 - 17.5 = 27$$

Interpretation

The middle 50% of commuting times span 27 minutes.

The Middle 50%



Key Point

The IQR ignores the lowest 25% and highest 25% of the data.

Percentiles

Definition

Percentiles divide a ranked data set into 100 equal parts.

$$P_k$$

means the kth percentile.

- P_{25} is the 25th percentile.
- P_{50} is the 50th percentile.
- P_{75} is the 75th percentile.

Connection

$$Q_1 = P_{25}, \qquad Q_2 = P_{50}, \qquad Q_3 = P_{75}$$

Finding a Percentile

Approximate Method

To find the kth percentile:

$$\text{Position} = \frac{k \times n}{100}$$

If the position is not a whole number, round up to the next whole number.

Important

The data must be ranked first.

Example: 70th Percentile

Use the commuting-time data:

$$7,\ 14,\ 17,\ 18,\ 24,\ 29,\ 37,\ 39,\ 42,\ 47,\ 55,\ 63$$

Find the 70th percentile.

$$k = 70, \qquad n = 12$$

$$\frac{k \times n}{100} = \frac{70 \times 12}{100} = 8.4$$

Round up:

$$8.4 \to 9$$

Conclusion

The 70th percentile is the 9th value, which is 42.

Percentile Rank

Definition

The percentile rank of a value gives the percentage of data values that are less than that value.

Formula:

$$\text{Percentile rank} = \frac{\text{Number of values less than } x}{\text{Total number of values}} \times 100\%$$

Example: Percentile Rank of 42

Ranked data:

$$7,\ 14,\ 17,\ 18,\ 24,\ 29,\ 37,\ 39,\ 42,\ 47,\ 55,\ 63$$

There are 8 values less than 42.

$$\text{Percentile rank of } 42 = \frac{8}{12} \times 100\%$$

$$= 66.67\%$$

Interpretation

About 67% of the students commute for less than 42 minutes.

Box-and-Whisker Plot

Definition

A **box-and-whisker plot** displays the center, spread, and skewness of a data set using five main values.

The five-number summary is:

$$\text{Minimum}, \quad Q_1, \quad \text{Median}, \quad Q_3, \quad \text{Maximum}$$

Why Use Boxplots?

Boxplots help us visualize spread, compare data sets, and detect possible outliers.

Five-Number Summary

For the commuting-time example:

$$7,\ 14,\ 17,\ 18,\ 24,\ 29,\ 37,\ 39,\ 42,\ 47,\ 55,\ 63$$

$$\text{Minimum} = 7$$

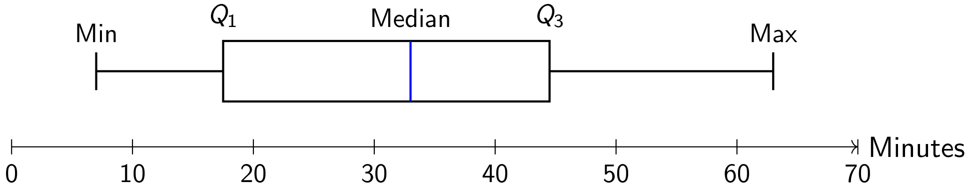
$$Q_1 = 17.5$$

$$\text{Median} = 33$$

$$Q_3 = 44.5$$

$$\text{Maximum} = 63$$

Simple Box-and-Whisker Plot



Interpretation

The box contains the middle 50% of the data.

Constructing a Boxplot: General Steps

1. Rank the data from smallest to largest.
2. Find Q_1, the median, and Q_3.
3. Compute the interquartile range:
$$IQR = Q_3 - Q_1$$
4. Find the inner fences:
$$Q_1 - 1.5(IQR), \qquad Q_3 + 1.5(IQR)$$
5. Draw the box, whiskers, and any outliers.

Inner Fences

Definition

Inner fences are boundaries used to detect possible outliers.

Lower inner fence:

$$Q_1 - 1.5(IQR)$$

Upper inner fence:

$$Q_3 + 1.5(IQR)$$

Outlier Rule

A value outside the inner fences is considered a possible outlier.

Example: Household Incomes

Income data, in thousands of dollars:

$$75,\ 69,\ 84,\ 112,\ 74,\ 104,\ 81,\ 90,\ 94,\ 144,\ 79,\ 98$$

Ranked data:

$$69,\ 74,\ 75,\ 79,\ 81,\ 84,\ 90,\ 94,\ 98,\ 104,\ 112,\ 144$$

$$Q_1 = 77, \qquad \text{Median} = 87, \qquad Q_3 = 101$$

$$IQR = 101 - 77 = 24$$

Finding the Inner Fences

$$IQR = 24$$

$$1.5(IQR) = 1.5(24) = 36$$

Lower inner fence:

$$Q_1 - 36 = 77 - 36 = 41$$

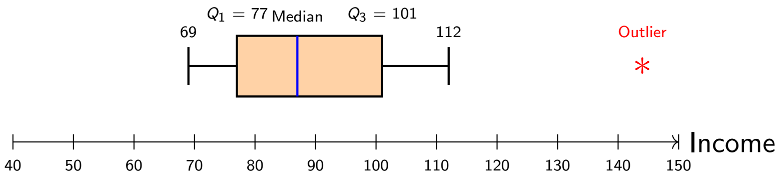
Upper inner fence:

$$Q_3 + 36 = 101 + 36 = 137$$

Conclusion

The value 144 is above the upper inner fence, so it is an outlier.

Boxplot with an Outlier



Important

The whisker does not extend to the outlier. It extends to the largest value within the inner fences.

Interpreting a Boxplot

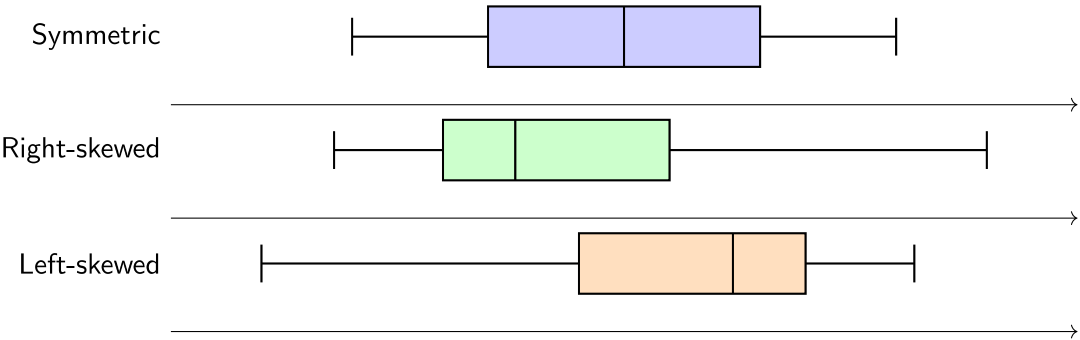
A boxplot can help us see:

- The center of the data.
- The spread of the data.
- Possible outliers.
- Whether the data are approximately symmetric or skewed.

Key Idea

The box shows the middle 50% of the data.

Skewness from Boxplots



Group Activity

Work in groups

The following data give the time, in minutes, that 20 students waited in line at a bookstore:

$$15,\ 8,\ 23,\ 21,\ 5,\ 17,\ 31,\ 22,\ 34,\ 6,$$

$$5,\ 10,\ 14,\ 17,\ 16,\ 25,\ 30,\ 3,\ 31,\ 19$$

Tasks:

1. Rank the data.
2. Find Q_1, the median, and Q_3.
3. Compute the IQR.
4. Find the five-number summary.
5. Construct a box-and-whisker plot.
6. Comment on the skewness.

Class Wrap-Up

Summary

- Quartiles divide ranked data into four equal parts.
- Q_2 is the median.
- $IQR = Q_3 - Q_1$.
- A boxplot uses the five-number summary.
- Inner fences help identify outliers.
- Boxplots help us see center, spread, outliers, and skewness.